

Problem Sheet 5

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1. In this question we work with first-order logic without equality; that is, we do not use the atomic formula $x = y$ where x, y are terms.

(a) Consider a signature τ containing only a binary relation symbol R . For each positive integer n show that there is a satisfiable τ -formula F_n such that every structure $\mathcal{A}$ that satisfies F_n has at least n elements.

(b) Fix a signature τ and τ -structures $\mathcal{A}, \mathcal{B}$ and assignments α, β . Consider a relation $\sim$ on pairs $(\mathcal{A}, \alpha), (\mathcal{B}, \beta)$ that satisfies the following two properties:

(P1) If $(\mathcal{A}, \alpha) \sim (\mathcal{B}, \beta)$ then for every atomic formula F we have $\mathcal{A} \models_{\alpha} F$ iff $\mathcal{B} \models_{\beta} F$.

(P2) If $(\mathcal{A}, \alpha) \sim (\mathcal{B}, \beta)$ then for each variable x we have (i) for each $a \in U^{\mathcal{A}}$ there exists $b \in U^{\mathcal{B}}$ such that $(\mathcal{A}, \alpha[x \mapsto a]) \sim (\mathcal{B}, \beta[x \mapsto b])$, and (ii) for all $b \in U^{\mathcal{B}}$ there exists $a \in U^{\mathcal{A}}$ such that $(\mathcal{A}, \alpha[x \mapsto a]) \sim (\mathcal{B}, \beta[x \mapsto b])$.

If $(\mathcal{A}, \alpha) \sim (\mathcal{B}, \beta)$, then show that for any formula F built from atomic formulae using the connectives $\neg, \wedge, \exists$, we have $\mathcal{A} \models_{\alpha} F$ iff $\mathcal{B} \models_{\beta} F$.

2. Are the following claims correct? Justify your answers.

(a) For any formula F and term t , if F is valid then $F[t/x]$ is valid.

(b) $\exists x (P(x) \rightarrow \forall y P(y))$ is valid.

(c) For any formula F and constant symbol c , if $F[c/x]$ is valid and c does not appear in F then $\forall x F$ is valid.

3. A group is a structure $(G, +, 0)$ where G is a set, $0 \in G$ is a special element called the identity and $+: G \times G \rightarrow G$ is a binary operation such that

(a) The operation $+$ is associative

(b) The constant 0 is a right-identity for the operation $+$

(c) Every element in G has a right inverse: for each $x \in G$, we can find $y \in G$ such that $x + y = 0$

(d) For any three elements $x, y, z \in G$, if $x + z = y + z$, then $x = y$

Using a signature $\tau = (c, \text{op})$ where c is a constant and op is a binary function symbol write (a)-(d) in FO.

4. Let τ be a signature consisting of the binary function symbol $+$ and a constant 0 . We denote by $x + y$ the function $+(x, y)$. Consider the following sentences:

$$\varphi_1 := \forall x \forall y \forall z [(x + (y + z)) = ((x + y) + z)]$$

$$\varphi_2 := \forall x [(x + 0) = x \wedge (0 + x) = x]$$

$$\varphi_3 := \forall x [\exists y (x + y = 0) \wedge \exists z (z + x = 0)]$$

Let ψ be the conjunction of the three sentences.

- (a) Show that ψ is satisfiable by exhibiting a τ -structure.
 - (b) Show that ψ is not valid.
 - (c) Let α be the sentence $\forall x \forall y ((x + y) = (y + x))$. Does α follow as a consequence of ψ ? That is, is it the case that $\psi \rightarrow \alpha$?
 - (d) Show that ψ is not equivalent to any of $\varphi_1 \wedge \varphi_2$, $\varphi_2 \wedge \varphi_3$ and $\varphi_1 \wedge \varphi_3$.
5. Show that the sentences $\forall x \exists y \forall z (E(x, y) \wedge E(x, z) \wedge E(y, z))$ and $\exists x \forall y \exists z (E(x, y) \wedge E(x, z) \wedge E(y, z))$ are not equivalent by exhibiting a graph which satisfies one but not both of the sentences.

6. For each $n \in \mathbb{N}$, $\exists^{\geq n}$ denotes a counting quantifier. Intuitively, $\exists^{\geq n}$ means that “there exist at least n such that”. FO with counting quantifiers is the logic obtained by adding these quantifiers (for each $n \in \mathbb{N}$) to the fixed symbols of FO. The syntax and semantics are as follows:

Syntax : For any formula φ of FO with counting quantifiers, $\exists^{\geq n} x \varphi$ is also a formula.

Semantics : $\mathcal{A} \models \exists^{\geq n} x \varphi$ iff $\mathcal{A} \models \varphi(a_i)$ for each of n distinct elements $a_1, a_2, \dots, a_n$ from the universe $u(\mathcal{A})$.

- (a) Using counting quantifiers, define a sentence φ_{45} such that $\mathcal{A} \models \varphi_{45}$ iff $|u(\mathcal{A})| = 45$.
 - (b) Define a FO sentence φ (not using counting quantifiers) that is equivalent to the sentence $\exists^{\geq n} x (x = x)$.
7. Write an FO formula that will evaluate to true only over a structure that has at least n elements and at most m elements.
8. Translate the following formula to rectified form, then to prenex form, and finally to Skolem form:

$$\forall z \exists y (Q(x, g(y), z) \vee \neg \forall x P(x)) \wedge \neg \forall z \exists x \neg R(f(x, z), z).$$